

ISP Testing

Чекори

1. Бираме карактеристики
 - Карактеристиките ги делиме на Interface based и Functionality based.
 2. Правиме партиции
 3. Проверуваме Disjoint, Complete
 4. Бираме Base Choice (happy path).
 5. Правиме Base Choice Coverage табела.
 6. Проверуваме дали некои тестови се Infeasible (невозможни).
 7. Тестираме
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Функција: `adx_signal(adx, plus_di, minus_di):`

1. Карактеристики

Interface-based:

C1: Any input value is NaN

b1 = True (at least one input is NaN)

b2 = False (all values are valid numbers)

C2: ADX strength

b1 = $adx \leq 25$ (weak/no trend)

b2 = $adx > 25$ (strong trend)

C3: Directional relationship (plus_di vs minus_di)

b1 = plus_di > minus_di (bullish)

b2 = plus_di == minus_di (neutral)

b3 = plus_di < minus_di (bearish)

Functionality-based:

C4: Signal returned

b1 = BUY

b2 = SELL

b3 = HOLD

2. Партиции

C1: {True, False}

C2: {<=25, >25}

C3: {plus_di>minus_di, plus_di==minus_di, plus_di<minus_di}

C4: {BUY, SELL, HOLD}

3. Disjoint & Complete

Disjoint

- Disjoint проверуваме во рамки на 1 карактеристика, не меду сите. Односно, земаме 1 карактеристика и ги гледаме нејзините партиции и притоа тие не треба да се преклопуваат, односно пресекот на партициите за конкретната карактеристика треба да биде празно множество.

C1: True $\cap$ False = $\emptyset$

C2: <=25 $\cap$ >25 = $\emptyset$

C3: three blocks do not overlap

C4: BUY $\cap$ SELL $\cap$ HOLD = $\emptyset$

Complete:

- Complete исто така проверуваме во рамки на 1 карактеристика. Сите партии на таа карактеристика треба да го покриваат целиот домен, односно унијата на сите партии да го даваат целиот домен.

C1: True U False = all input states

C2: ≤ 25 U > 25 = all ADX values

C3: $>$, $=$, $<$ covers all directional relationships

C4: BUY U SELL U HOLD = all possible signals

4. **Base Choice** -> C1=False, C2= >25 , C3=plus_di>minus_di, C4=BUY

5. Base Choice Coverage Table

Test	C1	C2	C3	C4	Feasible?
TC1	False	>25	plus_di > minus_di	BUY	BASE
TC2	True	>25	plus_di > minus_di	BUY	INFEASIBLE
TC3	False	≤ 25	plus_di > minus_di	BUY	INFEASIBLE
TC4	False	>25	plus_di == minus_di	BUY	INFEASIBLE
TC5	False	>25	plus_di < minus_di	BUY	INFEASIBLE
TC6	False	>25	plus_di > minus_di	SELL	INFEASIBLE
TC7	False	>25	plus_di > minus_di	HOLD	INFEASIBLE

6. Check infeasible tests

TC2: C1=True $\rightarrow$ C4 wrong (BUY $\rightarrow$ HOLD) -> feasible

TC3: C2= ≤ 25 $\rightarrow$ C4 wrong (BUY $\rightarrow$ HOLD) -> feasible

TC4: C3=equal → C4 wrong (BUY→HOLD) -> feasible

TC5: C3=bearish → C4 wrong (BUY→SELL) -> feasible

TC6: Тестот станува feasible ако се промени C4: SELL -> BUY, а тогаш тестот станува ист како и првиот па затоа го отфрламе.

TC7: Тестот станува feasible ако се промени C4: HOLD-> BUY, а тогаш тестот станува ист како и првиот па затоа го отфрламе.

7. Тестови

TC1	INPUT adx = 30 plus_di = 25 minus_di = 15	OUTPUT "BUY"
TC2	INPUT adx = NaN plus_di = 25 minus_di = 15	OUTPUT "HOLD"
TC3	INPUT adx = 20 plus_di = 25 minus_di = 15	OUTPUT "HOLD"
TC4	INPUT adx = 30 plus_di = 20 minus_di = 20	OUTPUT "HOLD"
TC5	INPUT adx = 30 plus_di = 15 minus_di = 25	OUTPUT "SELL"

Функција: `bollinger_signal(price, lower, upper):`

1. Карактеристики

Interface-based:

C1: Any input value is NaN

b1 = True (price, lower, or upper is NaN)

b2 = False (all values are valid numbers)

C2: Price position relative to Bollinger Bands

b1 = price < lower (below lower band)

b2 = lower <= price <= upper (inside bands)

b3 = price > upper (above upper band)

C3: Band configuration validity

b1 = True (lower < upper — valid bands)

b2 = False (lower >= upper — invalid bands)

Functionality-based:

C4: Signal returned

b1 = BUY

b2 = SELL

b3 = HOLD

2. Партиции

C1: {True, False}

C2: {price<lower, lower<=price<=upper, price>upper}

C3: {True, False}

C4: {BUY, SELL, HOLD}

3. Disjoint & Complete

Disjoint:

C1: True $\cap$ False = $\emptyset$

C2: three price positions do not overlap

C3: True $\cap$ False = $\emptyset$

C4: BUY $\cap$ SELL $\cap$ HOLD = $\emptyset$

Complete:

C1: True $\cup$ False = all input states

C2: below $\cup$ inside $\cup$ above = all price positions

C3: True $\cup$ False = all band configurations

C4: BUY $\cup$ SELL $\cup$ HOLD = all possible signals

4. Base Choice -> C1=False, C2=inside bands, C3=True, C4=HOLD

5. Base Choice Coverage Table

Test	C1	C2	C3	C4	Feasible?
TC1	False	C2.b1	C3.b1	BUY	BASE
TC2	True	C2.b1	C3.b1	BUY	INFEASIBLE
TC3	False	C2.b2	C3.b1	BUY	INFEASIBLE
TC4	False	C2.b3	C3.b1	BUY	INFEASIBLE
TC5	False	C2.b1	C3.b2	BUY	INFEASIBLE
TC6	False	C2.b1	C3.b1	SELL	INFEASIBLE
TC7	False	C2.b1	C3.b1	HOLD	INFEASIBLE

6. Check infeasible tests

TC2: C1=True → C4 wrong (BUY→HOLD) -> feasible

TC3: C4= HOLD→BUY -> feasible

TC4: C4=BUY -> SELL -> feasible

TC5: C5: BUY -> HOLD -> feasible

TC6: Тестот станува feasible ако се промени C4: SELL -> BUY, а тогаш тестот станува ист како и првиот па затоа го отфрламе.

TC7: Тестот станува feasible ако се промени C4: HOLD-> BUY, а тогаш тестот станува ист како и првиот па затоа го отфрламе.

7. Тестови

TC1	INPUT price = 30 lower = 40 upper = 60	OUTPUT "BUY"
TC2	INPUT price = NaN lower = 40 upper = 60	OUTPUT "HOLD"
TC3	INPUT price = 30 lower = 40 upper = 60	OUTPUT "BUY"

TC4	INPUT price = 70 lower = 40 upper = 60	OUTPUT "SELL"
TC5	INPUT price = 50 lower = 50 upper = 50	OUTPUT "HOLD"

Функција: `ma_signal(price, ma)`

1. Карактеристики

Interface-based:

C1: Any input value is NaN

b1 = True (price or ma is NaN)

b2 = False (both values are valid numbers)

C2: Price position relative to Moving Average

b1 = price > ma (price above MA - bullish)

b2 = price == ma (price equal to MA - neutral)

b3 = price < ma (price below MA - bearish)

C3: Price value

b1 = price > 0 (valid positive price)

b2 = price == 0 (zero price)

b3 = price < 0 (negative price)

Functionality-based:

C4: Signal returned

b1 = BUY

b2 = SELL

b3 = HOLD

2. Партиции

C1: {True, False}

C2: {price>ma, price==ma, price<ma}

C3: {price>0, price==0, price<0}

C4: {BUY, SELL, HOLD}

3. Disjoint & Complete

Disjoint:

C1: True $\cap$ False = $\emptyset$

C2: >, ==, < do not overlap

C3: >0, ==0, <0 do not overlap

C4: BUY $\cap$ SELL $\cap$ HOLD = $\emptyset$

Complete:

C1: True $\cup$ False = all input states

C2: >, ==, < covers all price-to-MA relationships

C3: >0 $\cup$ ==0 $\cup$ <0 = all numeric price values

C4: BUY $\cup$ SELL $\cup$ HOLD = all possible signals

4. Base Choice -> C1=False, C2=price>ma, C3=price>0, C4=BUY

5. Base Choice Coverage Table

Test	C1	C2	C3	C4	Feasible?
TC1	False	C2.b1	C3.b1	BUY	BASE
TC2	True	C2.b1	C3.b1	BUY	INFEASIBLE
TC3	False	C2.b2	C3.b1	BUY	INFEASIBLE
TC4	False	C2.b3	C3.b1	BUY	INFEASIBLE
TC5	False	C2.b1	C3.b2	BUY	FEASIBLE
TC6	False	C2.b1	C3.b3	BUY	FEASIBLE
TC7	False	C2.b1	C3.b1	SELL	INFEASIBLE
TC8	False	C2.b1	C3.b1	HOLD	INFEASIBLE

6. Check infeasible tests

TC2: C4: BUY→HOLD → feasible

TC3: C4: BUY→HOLD→ feasible

TC4: C4: BUY→SELL → feasible

TC7: Тестот станува feasible ако се промени C4: SELL-> BUY, а тогаш тестот станува ист како и првиот па затоа го отфрламе.

TC8: Тестот станува feasible ако се промени C4: HOLD-> BUY, а тогаш тестот станува ист како и првиот па затоа го отфрламе.

8. Тестови

TC1

INPUT
price=55
ma=50

OUTPUT
"BUY"

TC2

INPUT
price=NaN
ma=50

OUTPUT
"HOLD"

TC3

INPUT
price=50
ma=50

OUTPUT
"HOLD"

TC4

INPUT
price=45
ma=50

OUTPUT
"SELL"

TC5

INPUT
price=0
ma=-5

OUTPUT
"BUY"

TC6

INPUT
price=-5
ma=-10

OUTPUT
"BUY"